

Exercise 2

① Returning total number of employees

≡ SELECT COUNT(*) AS total-employees
FROM employees;

total-employees
10

② Returning total salary paid to all employees in the IT department

≡ SELECT SUM(salary) AS total-it-salary
FROM employees
WHERE department = 'IT';

total-it-salary
220 000

③ Calculating AVG(salary) of employees in the HR department

≡ SELECT AVG(salary) AS average-hr-salary
FROM employees
WHERE department = 'HR';

average-hr-salary
47500

④ Finding the highest and lowest salary in the company.

≡ SELECT MAX(salary) AS highest_salary,
MIN(salary) AS lowest_salary
FROM employees;

highest_salary	lowest_salary
62000	48000

⑤ Grouping employees by department and displaying the total salary paid in each department.

≡ SELECT department,
SUM(salary) AS total_salary
FROM employees

GROUP BY department;

Department	total_salary
Finance	119000
HR	99000
IT	220000
Marketing	105000

⑥ Counting how many employees work in each city (using GROUP BY and COUNT())

```
SELECT city,  
        COUNT(*) AS employees_count  
FROM employees  
GROUP BY city;
```

city	employees_count
Chicago	3
Houston	1
Los Angeles	2
New York	2
San Francisco	2

⑦ GROUPING employees by department calculating average salary and ordering by average salary descending

```
SELECT department,  
        AVG(salary) AS average_salary  
FROM employees  
GROUP BY department  
ORDER BY average_salary DESC;
```

department	average_salary
Finance	59500
IT	55000
Marketing	52500
HR	49500

⑤ Finding departments where the total salary paid exceeds 100000 using GROUP BY and HAVING

```

= SELECT department,
      SUM(salary) AS total_salary
FROM employees
GROUP BY department
HAVING SUM(salary) > 100000;

```

department	total_salary
Finance	119000
IT	220000
Marketing	105000

⑨ Returning Cities with more than one employee, ordered by count (using GROUP BY, HAVING, and ORDER BY)

```
SELECT city city, COUNT(*) AS employee-count  
FROM employees  
GROUP BY city  
HAVING COUNT(*) > 1  
ORDER BY employee-count DESC;
```

city	employee-count
Chicago	3
Los Angeles	2
New York	2
San Francisco	2

```
SELECT MAX(avg-salary) AS max-avg-salary  
FROM (SELECT AVG(salary) AS avg-salary  
FROM employees  
GROUP BY department)
```

AS dept-avg;

max-avg-salary
59500